

PCR Protocol

This protocol outlines the use of polymerase chain reaction (PCR) with Q5 high-fidelity polymerase and a thermocycler to amplify DNA. PCR is a fundamental technique in molecular cloning, enabling the exponential amplification of specific DNA sequences for downstream applications such as cloning and analysis. The use of Q5 polymerase ensures high accuracy and low error rates during DNA replication.

Note: This protocol depends on if a 25 or 50 uL is done.

Reagents

✓	Name	Quantity (for 25 uL reaction)	Quantity (for 50 uL reaction)	Final desired Concen tration	Location/Notes
☐	Q5 High-Fidelity 2X Master Mix	12.5 µL	25 µL	1x	
☐	10 µM Forward Primer	1.25 µL	2.5 µL	0.5 µM	
☐	10 µM Reverse Primer	1.25 µL	2.5 µL	0.5 µM	
☐	Template DNA	variable	variable	<1 000 ng	
☐	Nuclease-Free Water	to 25 µL	to 50 µL		

Equipment (+ Materials)

✓	Name	Quantity	Location/Notes
☐	PCR Thermocycler		
☐	Ice bucket		
☐	PCR Tube Holder		
☐	PCR Tubes		

Instructions

[Preparing PCR tubes]

1. Place well plate into the ice bucket to hold the PCR tubes
2. Label PCR tubes
3. Pipette desired primers, template DNA, and master mix into each PCR tube
 - a. Follow reagent quantities in table below:

Component	25 μ L reaction	50 μ L reaction	Final Concentration
Q5 High-Fidelity 2X Master Mix	12.5 μ L	25 μ L	1x
10 μ M Forward Primer	1.25 μ L	2.5 μ L	0.5 μ M
10 μ M Reverse Primer	1.25 μ L	2.5 μ L	0.5 μ M
Template DNA	variable	variable	<1 000 ng
Nuclease-Free Water	to 25 μ L	to 50 μ L	

- b. Consider a negative control without template DNA (adding water to make up volume) and a positive control with known template DNA and primer
4. Gently mix times, do not introduce bubbles by setting the micropipette to half of the volume of the master mix
5. Cap the PCR tubes and place them in the thermocycler
6. Start PCR using set program described below
7. Once program is finished, remove and store at 4°C

[Thermocycler Program]

STEP	TEMP	TIME
Initial Denaturation	98°C	30 seconds
25–35 Cycles	98°C	5–10 seconds
	50–72°C*	10–30 seconds
	72°C	20–30 seconds/kb
Final Extension	72°C	2 minutes
Hold	4–10°C	

*Note annealing temperature for Q5 is $T_m + 3^{\circ}\text{C}$

References

<https://www.neb.com/en-ca/protocols/protocol-for-q5-high-fidelity-2x-master-mix-m0492?srsId=AfmBOoodXI7pHW5C45AUqo4fpikgti470ETCXLrG7wEoANTgF9IGXXbk#>